

A Finite-Anchor Barrier for Physical Scale Laws Bounded Smooth Continuations and a Five-Scale Envelope Audit

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Abstract

The recent finite-memory route produces increasingly complete support certificates on five physical scales, but an all-level theorem requires control as the scale tends to zero. We place the available capacity, exterior-concentration, fourth-volume, fourth-mode, and memory-depth data into one aligned envelope ledger. Descriptive power-law fits are unstable: the median-capacity exponent is -4.496 , while leave-one-out fits range from -5.836 to -2.621 ; the worst fitted envelope misses an anchor by a factor of 142. We then prove a finite-anchor asymptotic nonidentifiability theorem. Any finite collection of positive anchors admits a smooth positive continuation with an arbitrary prescribed positive germ near zero. A bounded version preserves a known interval such as $0 < f < 1$ while producing continuations with limits 0, an arbitrary interior constant, or 1. Thus even exact finite anchors plus the natural capacity range cannot imply an all-level law. The negative fitted exponents of bounded ratios are also necessarily pre-asymptotic. An independent physical inequality, not extrapolation, is the next required input.

1 The five-scale question

The fourth-mode factorization is

$$q_4 = \frac{\nu_4}{\Lambda_{23}}, \quad \Lambda_{23} = q_2 q_3,$$

where $q_j = s_j/s_1$ and $\nu_4 = q_2 q_3 q_4$. RH-110 bounded the capacity Λ_{23} , RH-111 refined the exterior-volume estimate by a physical concentration factor, and RH-116 made memory depth an exact first-passage optimization. These are finite operator statements. To close the route at all later levels one would need scale-uniform bounds for the numerator, denominator, and directional tail factors.

It is tempting to fit the five observed scales and extrapolate. This paper separates two logically different claims:

1. a descriptive fit summarizes the finite archive;
2. an asymptotic law constrains every sufficiently small scale.

Exploratory summaries are useful [Tukey, 1977], but the first claim alone cannot imply the second.

2 Smooth finite-anchor extensions

Let $0 < \sigma_m < \dots < \sigma_1$ be distinct anchors and $y_i > 0$. Let g be any positive smooth function on a neighborhood $(0, a)$ of zero.

Theorem 1 (Positive finite-anchor extension). *There exists $f \in C^\infty((0, \sigma_1])$ such that*

$$f(\sigma_i) = y_i \quad (1 \leq i \leq m), \quad f(\sigma) = g(\sigma) \quad \text{for all sufficiently small } \sigma,$$

and $f(\sigma) > 0$ everywhere.

Proof. Choose $0 < a < b < \sigma_m$ and a smooth cutoff χ that is zero on $(0, a]$ and one on $[b, \sigma_1]$; standard bump functions provide such a cutoff [Hirsch, 1976]. Let p be the polynomial satisfying $p(\sigma_i) = \log y_i$. Define

$$f(\sigma) = \exp(\chi(\sigma)p(\sigma) + (1 - \chi(\sigma)) \log g(\sigma)).$$

The exponential is positive and smooth. At every anchor $\chi = 1$, while near zero $\chi = 0$, proving both identities. $\square$

The positivity construction may ignore a known physical range. The next version does not. For $A < B$, define the smooth diffeomorphism

$$T_{A,B}(y) = \log \frac{y - A}{B - y}, \quad A < y < B.$$

Theorem 2 (Bounded finite-anchor extension). *Suppose $A < y_i < B$ and $A < g(\sigma) < B$. There is a smooth extension matching all anchors and the germ as in Theorem 1, with $A < f(\sigma) < B$ at every scale.*

Proof. Interpolate the transformed anchor values $T_{A,B}(y_i)$ by a polynomial p . Blend p with $T_{A,B} \circ g$ using the same cutoff, and apply $T_{A,B}^{-1}$. The inverse maps all real values strictly into (A, B) and preserves both matching regions. $\square$

Corollary 3 (Asymptotic nonidentifiability). *For any finite capacity anchors in $(0, 1)$ there are smooth matching continuations, all remaining in $(0, 1)$, whose limits at zero are respectively 0, any chosen $c \in (0, 1)$, and 1.*

Proof. Apply Theorem 2 with germs $g_0(\sigma) = \sigma$, $g_c(\sigma) = c$, and $g_1(\sigma) = 1 - \sigma$ after restricting to scales below one. $\square$

Thus a finite data set cannot determine even the limiting value, much less a rate. The result is conditional only on the absence of an additional physical relation between scales; such a relation is exactly what the route must now prove.

Proposition 4 (Bounded negative-exponent obstruction). *If $0 < f(\sigma) \leq B < \infty$ and $f(\sigma) \sim c\sigma^\alpha$ as $\sigma \downarrow 0$ for $c > 0$, then $\alpha \geq 0$.*

Proof. If $\alpha < 0$, then $c\sigma^\alpha \rightarrow \infty$, contradicting the finite upper bound and asymptotic equivalence. $\square$

In particular, negative fitted exponents for Λ_{23} or q_4 , both bounded by one, cannot persist as genuine zero-scale power laws.

3 Aligned physical envelope audit

We align RH-110 capacity and volume records, RH-111 concentration records, and RH-116 depth records by scale, side, and time. To avoid triple counting threshold-dependent packet chains, the envelope table uses the primary 10^{-8} chain. It contains 120 physical updates.

σ	records	median Λ_{23}	median concentration	median q_4	max depth
0.16	8	$5.51 \cdot 10^{-8}$	1.0000	$1.21 \cdot 10^{-8}$	3
0.08	12	$2.28 \cdot 10^{-4}$	1.0030	$4.59 \cdot 10^{-5}$	5
0.04	22	$7.76 \cdot 10^{-4}$	1.0090	$4.58 \cdot 10^{-4}$	6
0.02	34	$2.63 \cdot 10^{-2}$	1.0001	$2.22 \cdot 10^{-2}$	4
0.01	44	$3.00 \cdot 10^{-2}$	1.0656	$2.70 \cdot 10^{-2}$	4

Table 1: Primary-chain medians and the largest first-passage depth.

Across all records, capacity ranges from 1.07×10^{-11} to 0.388, and concentration ranges from 1 to 2.330. The observed depth sequence 3, 5, 6, 4, 4 is bounded on the archive but not monotone in scale.

For each positive envelope statistic we fit $y(\sigma) = C\sigma^\alpha$ by least squares in log coordinates. These fits are explicitly descriptive, not inferential or asymptotic.

metric	exponent	R^2	max residual factor	leave-one-out exponent range
capacity minimum	-5.114	0.728	141.94	$[-6.561, -2.083]$
capacity median	-4.496	0.838	13.61	$[-5.836, -2.621]$
concentration median	-0.0179	0.517	1.028	$[-0.0249, -0.0009]$
fourth-mode median	-5.109	0.875	11.94	$[-6.573, -3.320]$

Table 2: Finite-anchor power-law diagnostics.

The apparently strong median trends coexist with order-of-magnitude residuals and large leave-one-out drift. Proposition 4 gives an even sharper warning: the negative exponents for bounded ratios must turn over or saturate before the asymptotic regime.

The archived constructive audit applies Theorem 2 to the five median-capacity anchors. Three continuations remain strictly in $(0, 1)$ and match every anchor to relative error below 1.7×10^{-15} . At probe scale 10^{-6} their values are 10^{-6} , $1/2$, and 0.999999.

4 Consequence and boundary

RH-117 turns a vague caution about extrapolation into a precise obstruction. The five-scale evidence is encouraging on the fine chain, but finite anchors cannot supply the missing all-level hypothesis, even after imposing the natural capacity range. RH-118 should therefore state and prove a conditional exterior-route theorem whose assumptions are explicit physical inequalities, then identify the smallest unresolved all-level inputs.

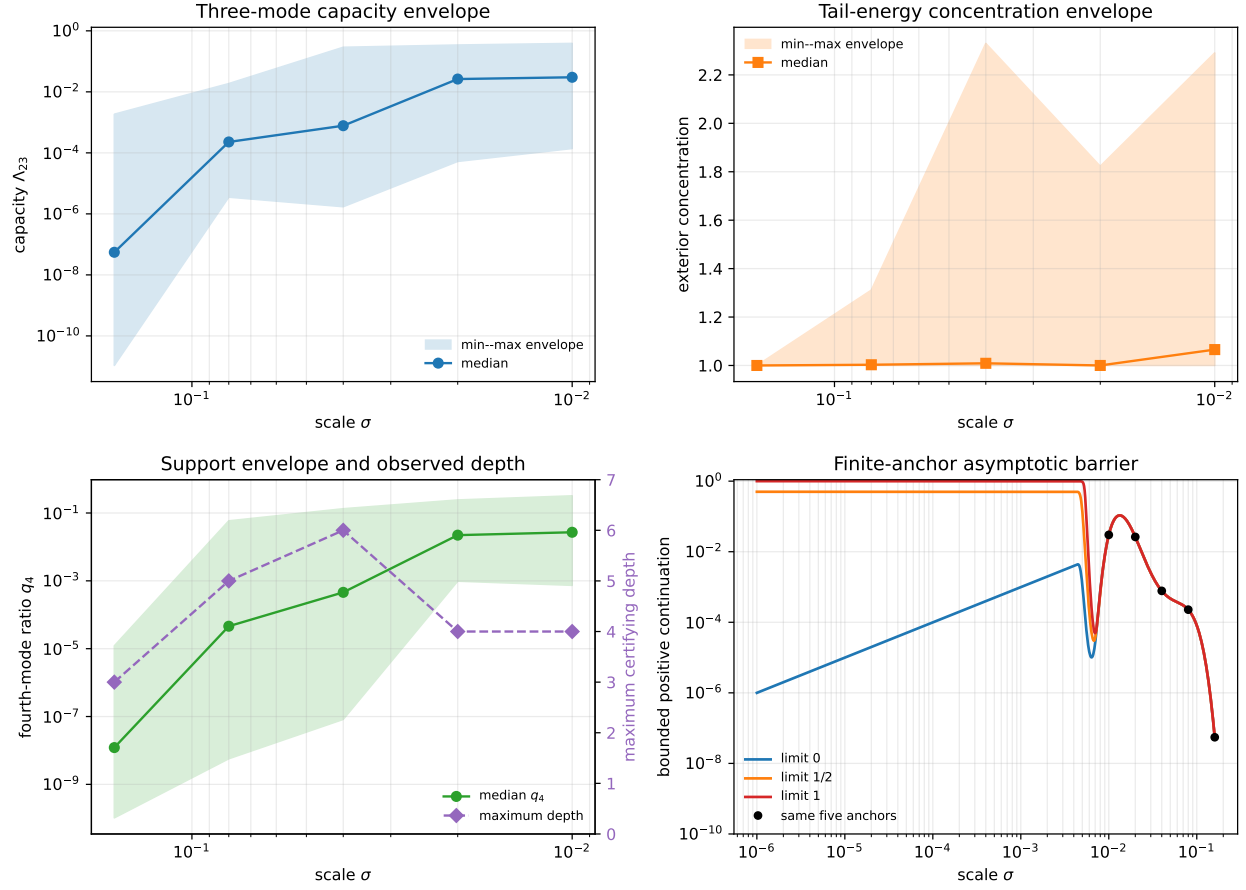


Figure 1: Physical envelopes, observed memory depth, and three bounded smooth continuations through the same capacity anchors.

This paper proves no all-level capacity, concentration, volume, directional, or bounded-depth law. It does not close uniform Stage A, construct a Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

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